

Indian Institute of Technology (Indian School of Mines) Dhanbad

Monsoon End Semester Question Paper

Examination: B.Tech (1st Year, Sections E, F, G, & H)

Time: 2 Hours

Discipline: Mechanical Engineering

Full Marks: 50

Subject: Manufacturing Process (NMES 101)

Session: 2025 - 26 (Monsoon)

(Answer all the questions)

Part – I (10 marks)

1. A weld is made using MIG welding process with the following parameters: current = 200 A, voltage = 25 volt, welding speed = 18 cm/min, diameter = 1.2 mm, and feed rate = 4 mm/min. Consider thermal efficiency is 65%. Determine the heat input in KJ/cm and the area of the cross section in mm². [2]
2. Spot welding of two 1.0 mm thick steel sheets (density = 8000 kg/m³) is carried out by passing certain amount of current for 0.1 second through the electrodes. The resultant weld nugget formed is 5 mm in diameter and 1.5 mm thickness. If the latent heat of fusion of steel is 1400 kJ/kg and the effective resistance in the welding operation is 200 Ω , calculate the current passing through the electrodes. [2]
3. In an arc welding process, the voltage and current are 25 V and 300 A respectively. The arc heat transfer efficiency is 0.85 and welding speed is 8 mm/sec. Determine the net heat input (in J/mm). [2]
4. Two metallic sheets, each of 2.0 mm thickness, are welded in a lap joint configuration by resistance spot welding at a welding current of 10 kA and welding time of 10 millisecond. A spherical fusion zone extending up to the full thickness of each sheet is formed. The properties of the metallic sheets are: Ambient temperature = 293 K, Melting temperature = 1793 K, Density = 7000 kg/m³, Latent heat of fusion = 300 kJ/kg, Specific heat = 800 J/kg K. Assume: (i) contact resistance along sheet-sheet interface is 500 micro-ohm and along electrode-sheet interface is zero; (ii) no conductive heat loss through the bulk sheet materials; and (iii) the complete weld fusion zone is at the melting temperature. Find the melting efficiency in the process. [2]
5. A direct current welding machine with a linear power source characteristic provides open circuit voltage of 80 V and short circuit current of 800 A. During welding using this machine, the measured arc current is found to be 500 A corresponding to an arc length of 5.0 mm and the measured arc current is found to be 460 A corresponding to an arc length of 7.0 mm. Determine the linear voltage (E)-arc length (L) characteristic of the welding arc (where E is in volt and L is in mm). [2]

Part – II (15 marks)

6. In Abrasive Water Jet Machining (AWJM), the material removal rate (MRR) is given by: [2+2]

$$MRR = u c_d \frac{\pi}{4} d_o^2 \sqrt{\frac{2p_w^3}{\rho_w}}$$

- (A) During cutting of a 6 mm thick titanium plate, AWJM system operates initially with: $u = 0.50$, $c_d = 0.84$, $d_o = 0.8$ mm, $p_w = 300$ MPa, $\rho_w = 1000$ kg/m³. It is known from experiments that the plate can be fully cut in a single pass only if the MRR is at least 40 mm³/s. Traverse speed of the jet is kept constant for all cases. (i) Calculate initial MRR for the given conditions. (ii) State whether the plate can be fully cut in a single pass under the initial conditions.

(B) Now the AWJM parameters are changed simultaneously as follows: Nozzle diameter is reduced by 25%, Water density is decreased to 0.82 times its original value due to heating, Abrasive mass flow rate is increased by 40% (assume this increases the constant u in the same proportion), Stand-off distance (SOD) is increased from 2 to 6 mm. Assume the effective water pressure at the workpiece is inversely proportional to SOD. Using the same MRR formula, determine (i) the new MRR, (ii) the percentage change in MRR with respect to the initial value, (iii) whether the plate can still be fully cut in a single pass at the same traverse speed. [3+2+1]

7. A stainless-steel alloy having following composition is electro-chemically machined (ECM) at a current of 1500 A with a current efficiency of only 88% due to side reactions ($F = 96500$). [2+3]

Element	Wt%	Atomic Wt	Valency
Fe	70%	55.85	2
Cr	20%	52.00	6
Ni	10%	58.70	2

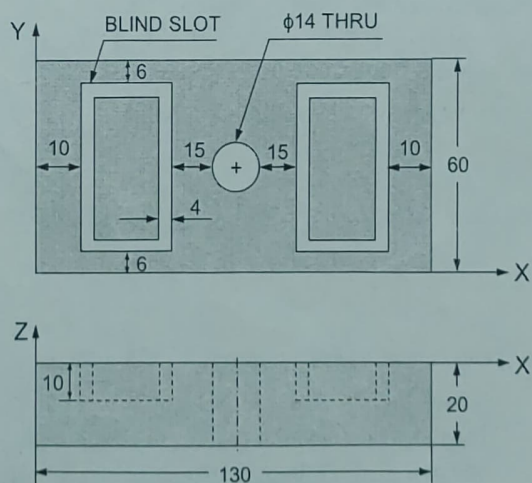
Compute the (i) electro-chemical equivalent E_{alloy} and (ii) actual MRR in g/s.

Part – III (15 marks)

8. Explain Merchant Circle diagram mentioning all forces and resultants as well as all angles. [4]
9. Write 10 parameters of the specification of the lathe machine. [2]
10. The cutting force and thrust force in an orthogonal cutting operation are 1470 N and 1589 N, respectively. Rake angle = 5° , width of the cut = 5.0 mm, chip thickness before the cut = 0.6 mm and chip thickness ratio = 0.38. Determine (a) the shear strength of the work material and (b) the coefficient of friction in the operation. [4]
11. In a turning operation on stainless steel, cutting speed = 150 m/min, feed = 0.25 mm/rev, and depth of cut = 7.5 mm. How much power will the lathe draw in performing this operation if its mechanical efficiency = 90%. Specific Energy of stainless steel is 2.8 N-mm/mm³. [1]
12. A lathe is used to turn a cylinder of 150 mm diameter for 500 mm length. Cutting speed = 2.50 m/s, feed = 0.30 mm/rev, and depth of cut = 3.0 mm. Determine (a) cutting time and (b) metal removal rate. [4]

Part-IV (10 marks)

13. Write a CNC G&M code program for cutting two identical rectangular blind slots and one through hole on a $130 \times 60 \times 20$ mm³ rectangular plate, as shown in the adjacent figure. Available end-milling tool (T05) has diameter of 4 mm and drill (T02) has diameter of 14 mm. [8]



14. Write two advantages and two disadvantages of the top-mounted robot over floor-mounted robot. [2]

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